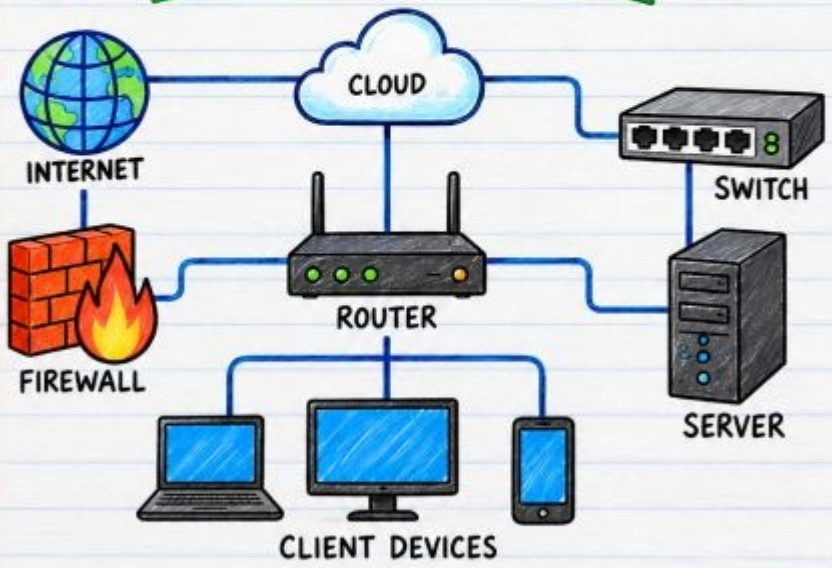
NETWORKING

COMPLETE NOTES

(Beginner to Advanced)



LEARN • UNDERSTAND • APPLY

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Introduction to Computer Networking

1 What is Networking?



Networking is the process of connecting two or more devices to exchange data and share resources using communication protocols.

2 Why Networking matters in DevOps

- ✓ Enables communication between tools, services & clouds.
- ✓ Critical for CI/CD, monitoring, logging & automation.
- ✓ Ensures security, performance, scalability & reliability.
- ✓ Foundation of Cloud, Containers & Microservices.



3 Client ↔ Server Architecture



- Client (User/Device) sends a request.
- Server processes the request.
- Server sends back the response.

4 LAN, WAN & Internet



LAN (Local Area Network)
- Covers a small area (home, office).
- Used to share files, printers, and devices.

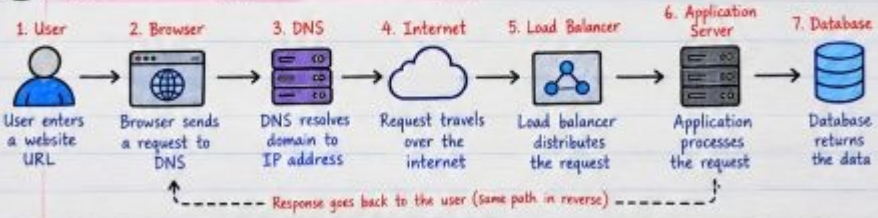


WAN (Wide Area Network)
- Covers a large geographic area (cities, countries).
- Connects multiple LANs.



Internet
- Global network of networks.
- Uses standard protocols (TCP/IP) to connect devices worldwide.

5 Real-world Request Flow (Example)



6 Where Networking is used in DevOps



Cloud Infrastructure	Containers & Orchestration	Microservices	CI/CD Pipelines	Monitoring & Logging	Security & Firewall Rules	Microservices Communication
Connect to cloud resources (AWS, Azure, GCP).	Enable container communication (Docker, Kubernetes).	Services communicate over network (REST, gRPC, Kafka).	Tools communicate (Git, Jenkins, SonarQube, etc.).	Collect and send metrics/logs (Prometheus, Grafana, ELK, Splunk).	Control network access using firewalls, security groups, and policies.	Internal & external service-to-service communication (API Gateway, Service Mesh).

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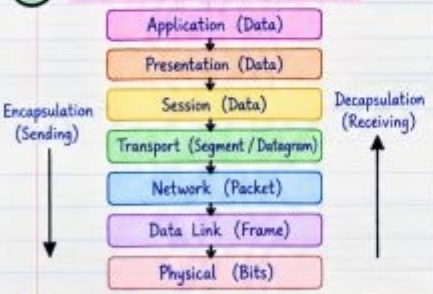
OSI Model (7 Layers)

The OSI (Open Systems Interconnection) model is a conceptual framework that standardizes how data is transmitted over a network.

1 7 Layers Overview

Layer (No.)	Layer Name	Responsibilities	Data Unit (PDU)	Devices at this Layer	Example / Protocols
7	Application Layer	Provides network services to user applications.	Data	–	HTTP, HTTPS, FTP, SMTP, DNS, SSH, DHCP
6	Presentation Layer	Handles data format, encryption, compression, translation.	Data	–	SSL/TLS, JPEG, ASCII, MPEG
5	Session Layer	Manages sessions, establishes, maintains and terminates connections.	Data	–	NetBIOS, RPC, PPTP
4	Transport Layer	Provides end-to-end delivery, segmentation, flow control and error control.	Segment (TCP) Datagram (UDP)	L4 Switch (Load Balancer)	TCP, UDP
3	Network Layer	Logical addressing, routing, path selection, packet forwarding.	Packet	Router (L3)	IP (IPv4, IPv6), ICMP, OSPF, BGP
2	Data Link Layer	Framing, MAC addressing, error detection and control.	Frame	Switch / Bridge (L2)	Ethernet, PPP, 802.11, MAC
1	Physical Layer	Transmits raw bits over the physical medium.	Bits	Hub, Repeater	Cables, Connectors, Hubs, Signal Standards

2 Data Unit (PDU) Flow



3 Real-world Example (Web Request)



1. Application Layer – User enters a URL (e.g., www.google.com)
2. Presentation Layer – Data is encrypted (HTTPS)
3. Session Layer – Session is established
4. Transport Layer – Data is broken into segments (TCP)
5. Network Layer – Packets are routed using IP
6. Data Link Layer – Frames are created with MAC address
7. Physical Layer – Bits are transmitted over the medium

4 Key Takeaways

- Upper 3 layers (5-7) focus on application communication.
- Middle layer (4) ensures reliable end-to-end delivery.
- Lower 2 layers (1-2) handle actual data transmission.
- Data flows down from Layer 7 → 1 when sending and up from 1 → 7 when receiving.

Remember

OSI model is a reference model.
Real networks use TCP/IP model in practice.

5 Memory Trick

Please Do Not Throw Sausage Pizza Away

- P – Physical
- D – Data Link
- N – Network
- T – Transport
- S – Session
- P – Presentation
- A – Application



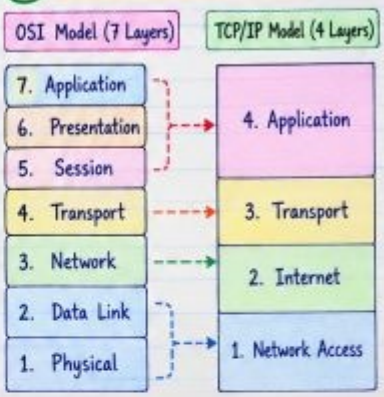
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TCP/IP Model (4 Layers)

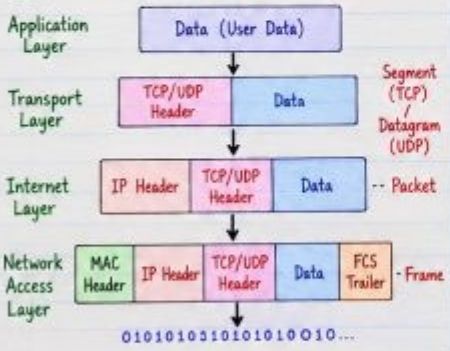
1 4 Layers of TCP/IP Model

Layer (Top to Bottom)		Key Responsibilities	Example Protocols
4	Application Layer	• Provides network services to user applications. • Handles user interface, APIs and data formats.	HTTP, HTTPS, FTP, DNS, SMTP, SNMP, DHCP
3	Transport Layer	• End-to-end communication. • Segmentation, reliability, flow control.	TCP, UDP
2	Internet Layer	• Logical addressing and routing. • Path selection between networks.	IP (IPv4, IPv6), ICMP, ARP
1	Network Access Layer	• Transmits data over the physical medium. • Framing, MAC addressing.	Ethernet, Wi-Fi, PPP, MAC

2 OSI vs TCP/IP Model



3 Encapsulation (At Sender)



4 Packet Journey (Sender to Receiver)



Packets may take different paths based on routing rules & network conditions.

Remember

Each layer adds its own header (and sometimes trailer) while sending, and removes it while receiving.

5 Why TCP/IP Model is Used?

- ✓ Simple and easy to understand.
- ✓ Widely adopted across the internet and all modern networks.
- ✓ Supports interoperability between different venders and platforms.
- ✓ Efficient, reliable and works well for large distributed systems.
- ✓ Forms the foundation for protocols used in DevOps tools, Cloud, containers and microservices.

Interview Tip

- Be clear about the role and function of each layer.
- Give real-life examples and devices.
- Also explain how data moves through encapsulation and decapsulation.
- Mention practical use cases (web browsing, file transfer, DNS, etc.).

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IP Addressing

1 IPv4 vs IPv6

IPv4	IPv6
• 32-bit address • Format: a.b.c.d (0 - 255 each octet) • Example: 192.168.1.10 • Total Addresses: ~ 4.3 Billion	• 128-bit address • Format: xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx • Example: 2001:0db8:85a3:8a2e:0370:7334 • Total Addresses: ~ 340 Undecillion

IPv6 was created to overcome IPv4 address exhaustion and provide better scalability & security.

2 Public vs Private IP

Public IP (Internet Reachable)



- Assigned by ISP
- Unique globally
- Used to communicate over the Internet
- Example: 8.8.8.8

Private IP (Internal Use)



- Used inside private networks (LAN / VPC)
- Not routable on the Internet
- Saves public IPs
- Examples:
  10.0.0.0/8 172.16.0.0/12
  192.168.0.0/16

3 Static vs Dynamic IP

Static IP



- Manually configured
- Does not change
- Used for servers, devices that need consistent access
- Example: 49.36.12.5

Dynamic IP

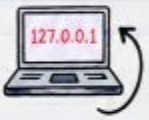
- Assigned automatically by DHCP
- Can change over time
- Used for client devices
- Example:



QA Insights Developer Laptop (Changes periodically)

4 Loopback Address

- Used by a device to communicate with itself.
- Used for testing and local development.
- IPv4 Loopback: 127.0.0.1
- IPv6 Loopback: ::1



Packets sent to loopback never leave the device.

5 CIDR (Classless Inter-Domain Routing) - Introduction

- CIDR is a method of allocating IP addresses and routing IP packets.
- It replaces traditional classful addressing.
- Notation: IP Address / Prefix Length
- Example: 192.168.1.0/24
  - 24 means first 24 bits are network part
  - Remaining 8 bits are for hosts
- CIDR helps in efficient IP usage and route summarization.

192.168.1.0/24



Common CIDR Ranges	
	/8 → 16,777,214 hosts
	/16 → 65,534 hosts
	/24 → 254 hosts
	/30 → 2 hosts (Point-to-Point)

Interview Tip

Understand IP types, ranges, and CIDR deeply - these are frequently asked in DevOps & Networking interviews!

Remember

IP addressing is the foundation of all networking. Without IP, devices cannot communicate!

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Subnetting & CIDR

1 Key Terms

- **Network ID** • Identifies the network portion of an IP address.
- **Host ID** • Identifies the specific device in that network.
- **Subnet Mask** • Divides an IP address into network and host portions.
- **CIDR Notation** • Short and efficient way to represent IP and subnet mask.
- **Available Hosts** • Total usable IP addresses in a subnet (excluding network & broadcast).

2 IP Address Structure (IPv4)



Example:

IP Address : 192.168.1.10
Subnet Mask : 255.255.255.0 (/24)

Binary View:

IP Address : 11000000 . 10101000 . 00000001 . 00001010
Subnet Mask : 11111111 . 11111111 . 11111111 . 00000000

Network ID (24 bits) | Host ID (8 bits)

3 Subnet Mask

A subnet mask is a 32-bit number that tells which part of the IP is Network ID and which part is Host ID.

Common Subnet Masks

CIDR	Subnet Mask	Network Bits	Host Bits
/8	255.0.0.0	8	24
/16	255.255.0.0	16	16
/24	255.255.255.0	24	8
/25	255.255.255.128	25	7
/26	255.255.255.192	26	6
/27	255.255.255.224	27	5
/28	255.255.255.240	28	4
/30	255.255.255.252	30	2

4 CIDR Notation

CIDR (Classless Inter-Domain Routing) removes the need for classful networks and allows variable length subnetting.

Format: IP Address / Prefix Length

Examples:

- 192.168.1.0/24 → First 24 bits are network portion
- 10.0.0.0/8 → First 8 bits are network portion
- 172.16.5.0/16 → First 16 bits are network portion

Interview Tip

CIDR helps in efficient IP allocation, route summarization and reduces routing table size in Routers.

5 Available Hosts Calculation

Formula:

Total Hosts = 2^n
Usable Hosts = $2^n - 2$

Where 'n' = number of host bits (We subtract 2 for Network ID and Broadcast ID)

Example: 192.168.1.0/24

Given IP : 192.168.1.0/24
Subnet Mask : 255.255.255.0

CIDR	/24
Host Bits (n)	8
Total Hosts (2^n)	2^8 = 256
Usable Hosts ($2^n - 2$)	256 - 2 = 254
Host Range	192.168.1.1 - 192.168.1.254
Network ID	192.168.1.0
Broadcast ID	192.168.1.255



Remember

- ✓ Network ID → First IP in the subnet (all host bits = 0)
- ✓ Broadcast ID → Last IP in the subnet (all host bits = 1)

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MAC Address & ARP

1 MAC Address (Media Access Control)

- A unique hardware address assigned to Network Interface Card (NIC).
- Works at Data Link Layer (Layer 2).
- 48-bit address (6 groups of 2 hex digits).
- Format: `00:1A:2B:3C:4D:5E`
- Uniquely identifies a device on a local network.
- Assigned by manufacturer (globally unique).



2 Physical vs Logical Address

Physical Address (MAC)

- Hardware address
- Layer 2 (Data Link)
- Assigned by Manufacturer
- Used for local network communication
- Permanent (not easily changed)

Example: `00:1A:2B:3C:4D:5E`

Logical Address (IP)

- Software address
- Layer 3 (Network)
- Assigned manually or by DHCP
- Used for end-to-end communication
- Can be changed

Example: `192.168.1.10`

3 ARP (Address Resolution Protocol)

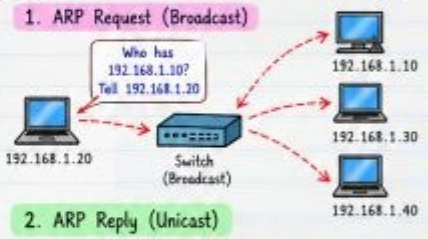
- ARP maps IP address (Logical) to MAC address (Physical).
- Works within the local network (broadcast domain).
- If sender doesn't know the MAC of destination IP, it sends an ARP Request (Broadcast).
- Destination replies with ARP Reply (Unicast) containing its MAC address.



Remember:
ARP is like asking: "Who has this IP? Tell me your MAC address."

4 ARP Process (Broadcast & Reply)

1. ARP Request (Broadcast)



2. ARP Reply (Unicast)



5 Switch MAC Table

- Switch learns MAC addresses from incoming frames.
- Stores entries in MAC Address Table with port information.
- Used to forward frames only to the correct port (not broadcast).

Example MAC Table	
MAC Address	Port
00:1A:2B:3C:4D:5E	Port 1
00:1A:2B:3C:4D:6F	Port 2
00:1A:2B:3C:4D:7A	Port 3
00:1A:2B:3C:4D:8B	Port 4
....	....

Key Points:

- ✓ MAC = Hardware Address (Layer 2)
- ✓ IP = Logical Address (Layer 3)
- ✓ ARP maps IP → MAC
- ✓ Switch uses MAC Table for intelligent forwarding

6 Packet Delivery Flow



1. Sender checks if destination IP is in same subnet.
2. If not, ARP is used to find destination MAC.
3. Sender sends frame to switch (with destination MAC).
4. Switch looks up MAC table and forwards to correct port.
5. Destination receives the frame and processes it.

Interview Tip:
Be clear about the difference between MAC & IP, how ARP works, and how switches learn & forward frames. These are commonly asked in DevOps, Network & Cloud interviews.

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TCP vs UDP

TCP and UDP are Transport Layer protocols used to transmit data between applications over an IP network.

1 Connection-Oriented vs Connectionless

TCP (Connection-Oriented)

- A connection is established before data transfer and closed after completion.
- Ensures reliable and ordered delivery of data.
- Used when data accuracy and completeness is important.

UDP (Connectionless)

- No connection is established.
- Data is sent directly without maintaining a session.
- No guarantee of delivery, order or duplicate protection.
- Used when speed and low latency is important.

3 Reliability vs Speed

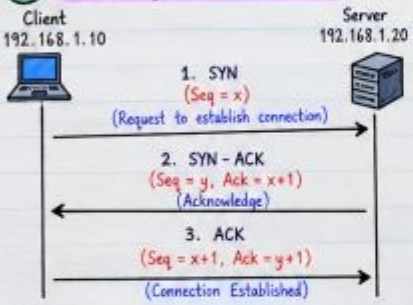
TCP

- ✓ Reliable delivery
- ✓ Ensures data arrives in order
- ✓ Error checking & retransmission
- ✓ Higher overhead
- ✓ Slower than UDP

UDP

- ✓ Best-effort delivery
- ✓ No guarantee of order or delivery
- ✓ Lower overhead
- ✓ Faster and lightweight
- ✓ Suitable for real-time applications

2 Three Way Handshake (TCP)



4 Real-World Examples

TCP (Reliable Apps)

- Web Browsing (HTTP/HTTPS)
- Email (SMTP, IMAP, POP3)
- File Transfer (FTP, SFTP)
- Database Connections (MySQL, PostgreSQL)

UDP (Fast Apps)

- Online Gaming
- Video Streaming (YouTube, Netflix)
- Voice over IP (VoIP) (Zoom, Teams)
- Live Streaming
- DNS Queries

5 TCP vs UDP Comparison

Feature	TCP	UDP
Connection	Connection-Oriented	Connectionless
Reliability	Reliable (ACK + Retransmission)	Not Reliable (Best Effort)
Data Order	Maintains Order	No Order Guarantee
Error Handling	Yes (Detection & Recovery)	No Error Recovery
Data Delivery	Guaranteed	Not Guaranteed
Speed	Slower	Faster
Header Size	Larger (20-60 bytes)	Smaller (8 bytes)
Use Case	Web, Email, File Transfer, DB, etc.	Gaming, Streaming, VoIP, DNS, etc.

6 Key Takeaways

- ✓ TCP is like a phone call (connection, reliable, ordered).
- ✓ UDP is like a courier (fast, no connection, no guarantee).
- ✓ Use TCP when accuracy matters.
- ✓ Use UDP when speed and low latency matters.

7 Interview Tip

Be ready to explain:

- When to use TCP vs UDP?
- How 3-way handshake works?
- Header size difference?
- Real-world examples and use cases.

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Ports & Common Protocols

Ports are logical endpoints used by protocols to identify applications and services on a host.

Protocol / Service	Port	Transport	Purpose
HTTP (HyperText Transfer Protocol)	80	TCP	Unencrypted web traffic (normal browsing)
HTTPS (HTTP Secure)	443	TCP	Encrypted web traffic (SSL/TLS)
SSH (Secure Shell)	22	TCP	Secure remote login & command execution
DNS (Domain Name System)	53	UDP/TCP	Domain name resolution (Converts domain to IP)
FTP (File Transfer Protocol)	21 (Control) 20 (Data)	TCP	File transfer (unencrypted)
SFTP (SSH File Transfer Protocol)	22	TCP	Secure file transfer over SSH
SMTP (Simple Mail Transfer Protocol)	25	TCP	Sending emails (Mail server)
Database Ports			
MySQL	3306	TCP	MySQL Database
PostgreSQL	5432	TCP	PostgreSQL Database
MongoDB	27017	TCP	MongoDB Database
Redis	6379	TCP	Redis Cache / Key-Value Store

Key Points

- ✓ Port + IP address uniquely identifies a service on a host.
- ✓ Well-known ports : 0 – 1023
  (Used for common services like HTTP, SSH, DNS)
- ✓ Registered ports : 1024 – 49151
  (Used for specific applications)
- ✓ Dynamic / Private ports : 49152 – 65535
  (Used for temporary / client connections)

Interview Tips

- Always mention the protocol along with the port (e.g., 80/TCP, 53/UDP).
- Know which services use TCP vs UDP.
- Be aware of firewall rules, as they are based on IP, port and protocol.
- Explain the difference between well-known, registered and dynamic ports.
- Commonly asked in DevOps, Cloud, Network and Security interviews.

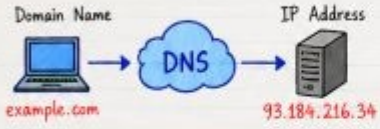
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DNS (Domain Name System)

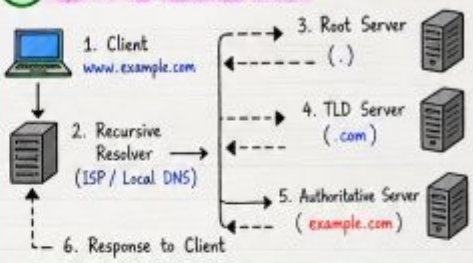
DNS is the Internet's phonebook. It translates human-readable domain names (like example.com) into IP addresses (like 93.184.216.34) that computers use to communicate.

1 What is DNS?

- DNS is the Internet's **phonebook**.
- It translates human-readable domain names into IP addresses that computers use to communicate.
- It works at the **Application layer**.



2 DNS Resolution Flow



Caching is used to speed up future lookups.

3 Important DNS Record Types

Record Type	Purpose	Example	Explanation
A (Address)	Maps a domain name to an IPv4 address.	example.com → 93.184.216.34	Points the domain to an IPv4 address.
CNAME (Canonical Name)	Points one domain name to another domain name.	www.example.com → example.com	Alias of a domain name to another.
MX (Mail Exchange)	Specifies mail server for the domain.	example.com → mail.example.com	Tells where to deliver emails for the domain.
TXT (Text)	Stores text information about the domain.	example.com → "v=spf1 include:_spf.example.com ~all"	Used for SPF, security verification, policies, etc.

4 DevOps Example (AWS Route 53)

Application hosted on AWS with DNS



Why DNS is Critical in DevOps?

- ✓ Helps access to cloud resources / services.
- ✓ Enables high availability using multiple records.
- ✓ Helps with CI/CD, monitoring & alerting.
- ✓ Used for service discovery in microservices.

5 Quick Reference (Common DNS Records)

Type	Use Case
A	Domain → IPv4 address
AAAA	Domain → IPv6 address
CNAME	Alias name → (points to another domain)
MX	Mail server configuration
TXT	SPF, DKIM, verification, policies
NS	Authoritative name server
PTR	Reverse DNS (IP → Domain)

6 Interview Tip

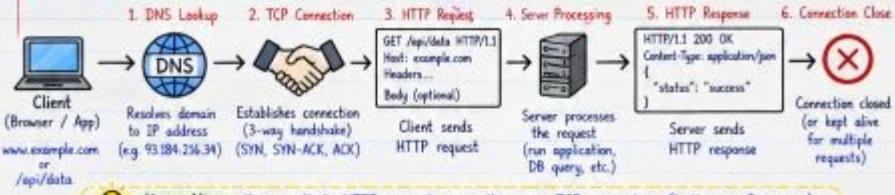
- Be ready to explain what DNS is, how it works, and the role of each DNS record type.
- ✓ Cover real-world examples (web hosting, email, load balancing, microservices, etc.).
- ✓ Also discuss caching, TTL, and how DNS helps in high availability and disaster recovery.

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HTTP & HTTPS

HTTP and HTTPS are application layer protocols used to transfer data between a client (browser/app) and a server over the internet.

1 Request Lifecycle (How HTTP Works)



Keep-Alive allows multiple HTTP requests over the same TCP connection (better performance).

2 HTTP Methods

Method	Purpose	Idempotent	Safe
GET	Retrieve data	✓	✓
POST	Create new resource	✗	✗
PUT	Update entire resource	✓	✗
PATCH	Update partial resource	✗	✗
DELETE	Delete resource	✓	✗
HEAD	Get headers only (no response body)	✓	✓
OPTIONS	Get supported methods (CORS)	✓	✓

3 HTTP Status Codes (Most Important)

Category	Code	Meaning	Example
1xx Informational	100	Continue	Request received, continue process
2xx Success	200	OK	Request successful
	201	Created	Resource created
3xx Redirection	301	Moved Permanently	URL permanently moved
	302	Found	Temporary redirect
4xx Server Error	400	Bad Request	Invalid request
	401	Unauthorized	Authentication required
	403	Forbidden	No permission
	404	Not Found	Resource not found
	500	Internal Server Error	Server error
	502	Bad Gateway	Invalid response from upstream server
	503	Service Unavailable	Server overloaded
	504	Gateway Timeout	Upstream server timeout

4 SSL/TLS (How HTTPS Works)



Key Points

- ✓ HTTPS uses TLS (Transport Layer Security).
- ✓ Certificates are used for authentication.
- ✓ Data is encrypted in transit.
- ✓ Default ports:
  - HTTP = 80
  - HTTPS = 443

5 REST APIs

- REST (Representational State Transfer) is an architectural style for APIs.
- Uses HTTP methods to perform operations on resources.
- Resources are identified using URLs.
- Common data format: JSON (also XML).

Examples:

```
GET    /api/users      → Get all users
POST   /api/users      → Create new user
GET    /api/users/123  → Get user by ID
PUT    /api/users/123  → Update user
DELETE /api/users/123  → Delete user
```

6 HTTPS Benefits

https://example.com

- ✓ Encrypts data (prevents eavesdropping).
- ✓ Authenticates server (prevents man-in-the-middle attacks).
- ✓ Ensures data integrity (cannot be tampered).
- ✓ Builds trust with users (secure website).
- ✓ Required for modern web apps, APIs, payments, logins.



7 Quick Reference (HTTP vs HTTPS)

Feature	HTTP (80)	HTTPS (443)
Data	Plain Text	Encrypted (TLS)
Security	No Security	Secure
Port	80	443
URL	http://	https://
Use Case	Non-sensitive data	Sensitive data (logins, payments)

Interview Tip

Be ready to explain the HTTP request/response lifecycle, difference between HTTP and HTTPS, status codes, methods, and real-world examples.

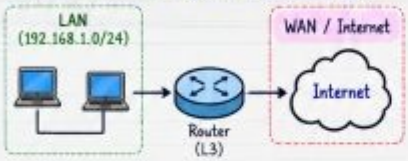
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Routing, NAT & Gateway

These are key networking concepts that help devices communicate within and outside a network.

1 Router (What it does)

- A router connects multiple networks and forwards packets based on IP address.
- It uses a routing table to decide the best path for the packet.
- Routers work at Layer 3 (Network Layer).



2 Routing (How packets find the path)

- Router uses routing table to find the next hop for a destination network.
- Each entry in the table contains the destination network and the next hop (or directly connected).

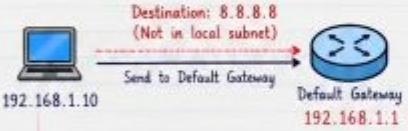
Routing Table (Example)

Destination Network	Next Hop	Interface
192.168.1.0/24	Directly Connected	LAN
10.0.0.0/24	10.0.0.1	eth0
0.0.0.0/0	203.0.113.1	eth1 (Internet)

Key Point: 0.0.0.0/0 is the Default Route. Used for all unknown destinations (Internet).

3 Default Gateway

- A default gateway is the router's IP address in your local network.
- If the destination IP is outside the local subnet, traffic is sent to the default gateway.



4 NAT (Network Address Translation)

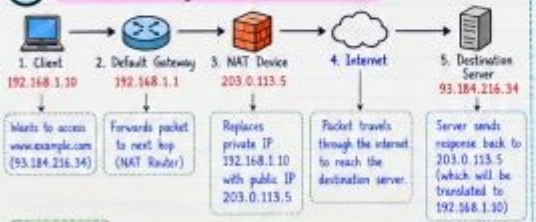
- NAT allows multiple private IPs to share one public IP address.
- It hides the internal private network from the internet.
- Used in home routers, cloud (AWS NAT Gateway), firewalls, etc.



Types of NAT:

- SNAT (Source NAT) → used for outgoing traffic (private → public)
- DNAT (Destination NAT / Port Forwarding) → used for incoming traffic (public → private)

5 Packet Journey (Inside → Internet)



Return Path:

Server → Internet → NAT (203.0.113.5) → 192.168.1.10 (Original Client)

6 Real-World Example (AWS VPC)

Application is hosted inside AWS VPC and accessed by users over the internet.



- Users access: https://example.com
- DNS resolves to 3.10.20.30 (Elastic IP)
- Traffic goes to NAT Gateway → routed to App Server (10.0.1.10)
- Response follows the same path.

Interview Tips

- ✓ Explain what a router does and how routing works.
- ✓ What is default gateway and when it is used?
- ✓ Explain NAT (SNAT, DNAT) with real examples.
- ✓ Understand packet flow (inside to internet and back).
- ✓ Be ready to explain route table and default route (0.0.0.0/0).

Key Takeaways

- ✓ Router = Connects networks & forwards packets (Layer 3).
- ✓ Default Gateway = Router IP for external communication.
- ✓ NAT = Translates private IPs to public IP (hides internal network).
- ✓ Routing Table = Decides the best path.
- ✓ 0.0.0.0/0 = Default route (Internet).

Common Commands (Linux)

Show routing table	: ip route
Check default gateway	: ip route | grep default
Test connectivity	: ping 8.8.8.8
Trace route	: traceroute example.com
View NAT rules	: iptables -t nat -L
Check public IP	: curl ifconfig.me

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Firewalls & Network Security

1 What is a Firewall?

- A firewall monitors and controls incoming and outgoing network traffic based on security rules.
- It acts as a barrier between trusted internal network and untrusted external network (Internet).



2 Stateful vs Stateless Firewall

Feature	Stateful Firewall (Stateful Inspection)	Stateless Firewall (Packet Filtering)
Tracks Connection	Yes (maintains state of active connections)	No (checks each packet individually)
Decision Basis	Uses Layer 3, 4 and connection state	Uses Layer 3 and 4 info only (IP, Port, Protocol)
Performance	Slower (more processing)	Faster
Security	More secure	Less secure
Use Case	Enterprise networks, Cloud (e.g., AWS Security Groups)	Simple networks, basic filtering (e.g., NACLs, routers, ACLs)

3 Security Groups (Stateful)

- Virtual firewall for instances (e.g., AWS EC2).
- Stateful: allows return traffic automatically.
- Applied at instance level.
- Supports allow rules only (no deny).
- Example: Allow SSH (22), HTTP (80) from specific IP.

Example Security Group Rules

Type	Protocol	Port	Source	Description
SSH	TCP	22	203.0.113.10	Admin access
HTTP	TCP	80	0.0.0.0/0	Web access
HTTPS	TCP	443	0.0.0.0/0	Secure web

Stateful → if you allow inbound traffic, return traffic is automatically allowed.

4 Network ACL (NACL) (Stateless)

- Operates at subnet level (e.g., AWS VPC).
- Stateless: you must allow both inbound and outbound traffic.
- Supports allow and deny rules.
- Rules are evaluated in order (lowest number first).
- Example: Allow HTTP (80), HTTPS (443) and deny all others.

Example NACL Rules (Inbound)

Rule #	Type	Protocol	Port	Source	Action
100	HTTP	TCP	80	0.0.0.0/0	ALLOW
110	HTTPS	TCP	443	0.0.0.0/0	ALLOW
120	All	All	All	0.0.0.0/0	DENY

NACL is stateless → you must explicitly allow return traffic. A final deny-all rule is recommended.

5 VPN (Virtual Private Network)

- Creates an encrypted tunnel over the Internet.
- Allows secure access to private network resources.
- Used for remote access (users) or site-to-site (office).
- Protects data from eavesdropping.
- Protocols: IPSec, OpenVPN, SSL/TLS.



6 Bastion Host (Jump Server)

- A secure entry point to access private instances.
- Usually placed in a public subnet.
- Admins connect to bastion host first, then access internal servers.
- Reduces direct public exposure of private instances.
- Best practice: Use SSH with key pairs and restrict access by IP.



Key Points

- ✓ Firewall controls and filters network traffic based on rules.
- ✓ Security Group = Stateful (allow only).
- ✓ NACL = Stateless (allow/deny, subnet level).
- ✓ VPN provides secure encrypted connection.
- ✓ Bastion host is a secure jump server.
- ✓ Use least privilege and restrict access by IP.

Interview Tips

- Explain stateful vs stateless firewall with real examples.
- Difference between Security Group and NACL.
- How return traffic works in stateful and stateless.
- When to use VPN and Bastion Host.
- Common use cases in AWS / Cloud environments with diagrams.

Common Commands (Linux)

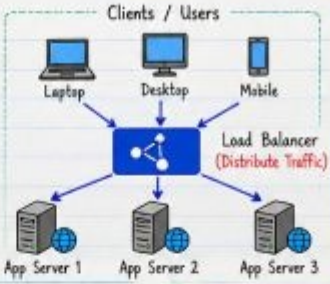
```
ip route                          # Show route table
ip route | grep default           # Check default gateway
ping 8.8.8.8                      # Test connectivity
traceroute example.com            # Trace route to host
curl ifconfig.me                  # Check public IP
netstat -tulnp                    # Check open ports
ssh -i key.pem ec2-user@<ip>      # SSH to server
```

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Load Balancer & Proxy

1 Why Load Balancer?

- Distributes incoming traffic across multiple servers.
- Improves availability, scalability, and performance.
- Prevents single point of failure.
- Performs health checks and routes traffic to healthy servers.



Key Point

- ✓ Load balancer helps handle more traffic without single server failure,
- ✓ improving reliability, scalability and performance.

2 L4 vs L7 Load Balancing

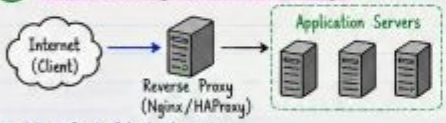
Feature	Layer 4 (Transport Layer)	Layer 7 (Application Layer)
Layer	L4 (TCP / UDP)	L7 (HTTP / HTTPS)
Decision	IP, Port, Protocol	Content (URL, Host, Header)
Speed	Very Fast	Slightly Slower
Example	NLB (AWS), TCP proxy	ALB (AWS), Nginx, HAProxy
Use Case	High performance, non-HTTP traffic	HTTP/HTTPS traffic, path-based routing, advanced rules

3 Forward Proxy (Client-side Proxy)



- Acts on behalf of client to access the internet.
- Hides internal IP from external websites.
- Used for caching, filtering, access control, etc.

4 Reverse Proxy (Server-side Proxy)



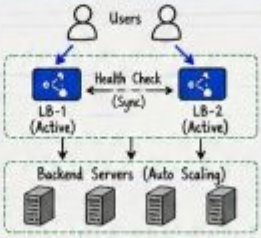
- Sits in front of backend servers.
- Receives client requests and forwards to appropriate server.
- Provides SSL termination, load balancing, caching.
- Hides backend servers from public.

5 Nginx (Popular Reverse Proxy / Load Balancer)

- High performance, lightweight and open-source.
- Used as reverse proxy, load balancer, cache.
- Load balances HTTP/HTTPS traffic.
- Supports L4 & L7 load balancing.
- Used for SSL termination, compression.
- Widely used in DevOps, Microservices, Kubernetes, Ingress Controllers.

6 High Availability (HA) Concepts

- Multiple load balancers (Active-Active).
- Health checks remove unhealthy servers.
- Auto scaling adds/removes servers.
- Use CDN or Anycast IP for global availability.
- Reduce single point of failure with multi-AZ setup.



Remember

- ✓ Use Health Checks for failure detection.
- ✓ Use Auto Scaling to handle traffic spikes.
- ✓ Deploy across multiple Availability Zones (AZ).
- ✓ Keep configuration simple and monitor performance.

Interview Tips

- Explain difference between L4 & L7 load balancer.
- When to use Nginx vs Load Balancer (ALB/NLB)?
- How health checks work?
- What happens if one server goes down?
- Difference between Forward Proxy and Reverse Proxy.
- How to achieve high availability?

Common Tools

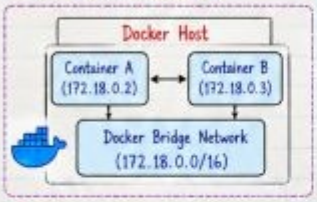
Nginx	Reverse Proxy, Load Balancer, Web Server
HAProxy	High performance L4/L7 Load Balancer
AWS ALB	Application Load Balancer (L7)
AWS NLB	Network Load Balancer (L4)
F5	Enterprise Load Balancer
Cloudflare	Reverse Proxy, CDN, DDoS Protection
Kubernetes	Ingress Controller, Service Load Balancing

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Docker Networking

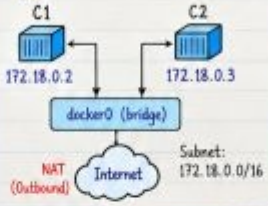
1 Docker Network Overview

- Docker provides multiple network drivers to enable containers to communicate with each other and external networks.
- Each container gets an IP address from the network subnet.
- We can create custom networks for better isolation and security.



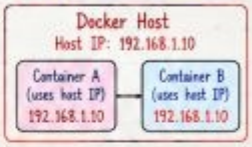
2 Bridge Network (Default)

- Default network mode.
- Containers connect to an isolated bridge (docker0) on the host.
- Containers can talk to each other using container IPs and can also access external networks through NAT.



3 Host Network

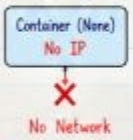
- Container shares the host's network namespace.
- No isolation – container gets host's IP.
- High performance, low latency.



Use Case:
High performance apps, monitoring, network tools.

4 None Network

- Container has no network interface.
- No IP address is assigned.
- Completely isolated container.



Use Case:
Security testing, sensitive workloads that should not access network.

5 Overlay Network

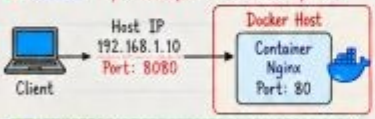
- Used for multi-host container communication.
- Commonly used in Docker Swarm / Kubernetes.
- Uses VXLAN under the hood.



Use Case:
Cross-host communication, multi-host environments (swarm/k8s).

6 Port Mapping (Publish Ports)

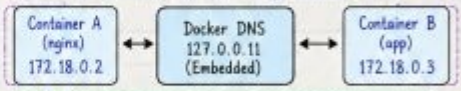
- Expose container port to the host.
- Format: `-p <host_port>:<container_port>`



Access:
http://192.168.1.10:8080 → container port 80
-p 8080:80 → maps host 8080 to container 80

7 Docker DNS (Embedded DNS)

- Docker provides an internal DNS server (127.0.0.11) for container name resolution.
- Containers can reach each other using container names.



Example:
In Container A, you can curl http://app:8080
('app' resolves to 172.18.0.3)

Key Points

- ✓ Docker has multiple network drivers (bridge, host, none, overlay, macvlan).
- ✓ Each container gets an IP from network subnet.
- ✓ Use custom networks for isolation.
- ✓ Bridge uses NAT for internet access.
- ✓ Host mode shares host network (no isolation).
- ✓ Overlay is used for multi-host communication.
- ✓ Embedded DNS (127.0.0.11) resolves container names.
- ✓ Port mapping exposes container ports to host.

Interview Tip

- ✓ Explain difference between all network drivers with use cases.
- ✓ When to use host vs bridge vs overlay?
- ✓ What is docker0?
- ✓ How container name resolution works in Docker?
- ✓ How port mapping works?
- ✓ Real-world scenario: multiple containers communicating across hosts.

Common Docker Network Commands

```
docker network ls                          # List networks
docker network create my-net               # Create network
docker network inspect my-net              # Inspect network
docker network rm my-net                   # Remove network
docker network prune                       # Remove unused
docker run -d --name web \
  --network my-net nginx                   # Run with custom net
docker run --network host nginx            # Host network
docker run --network none nginx            # None network
docker run -p 8080:80 nginx                # Port mapping
docker exec -it <container> bash           # Test inside
```

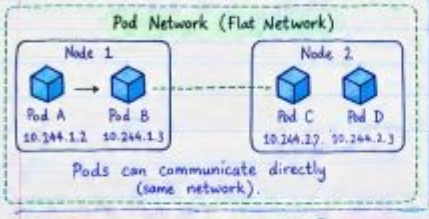
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Kubernetes Networking

1 Pod Network

- Each Pod gets its own IP address.
- Pods can communicate with each other across nodes without NAT.
- Implemented using CNI plugins (Flannel, Calico, Cilium, Weave).

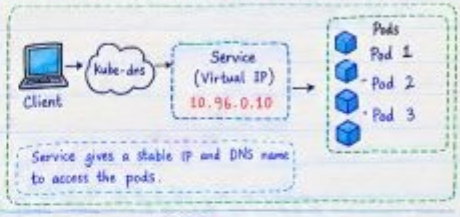
Key Point
Every Pod IP is reachable from every other Pod in the cluster.



2 Service (Overview)

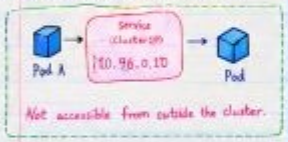
- Service provides a stable endpoint (IP + DNS) to a set of Pods.
- Decouples clients from Pod lifecycle.
- Kubernetes Service Types:
  ClusterIP | NodePort | Load Balancer | Ingress

Key Point
- Service provides stable access to pods (even if pods change).



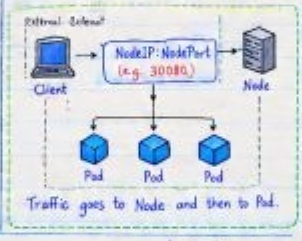
3 ClusterIP (Default)

- Exposes the service on a cluster-internal IP.
- Not accessible from outside the cluster.
- Used for internal communication.



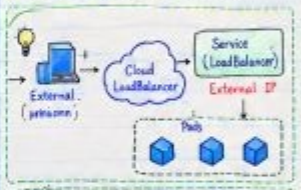
4 NodePort

- Exposes the service on each node's IP at a static port (30000–32767).
- Accessible from outside via <NodeIP>:<NodePort>.



5 Load Balancer

- Creates an external Load Balancer in cloud (AWS, EBS, Azure LB, GCP LB).
- Provides a dedicated external IP.



Key Point
Load balancer with a public IP.

6 Ingress

- Provides HTTP/HTTPS routing to services.
- Works at Layer 7 (Application layer).
- Requires Ingress Controller (e.g., Nginx Ingress).

Key Point
Ingress works at Layer 7 and needs an Ingress Controller. It provides host-based and path-based routing.



7 Service Types Summary

Type	Exposure	Access	Use Case
ClusterIP	Internal IP	Inside Cluster	Internal services
NodePort	Node IP + Port	Inside & Outside	Dev / Testing
LoadBalancer	Cloud LB (External IP)	Outside	Production Apps
Ingress	HTTP/HTTPS (L7)	Outside (via rules)	Web Apps / APIs

Interview Tips

- ✓ Explain difference between L4 & L7.
- ✓ Understand when to use NodePort vs LoadBalancer.
- ✓ Be ready with examples (Nginx, HAProxy).
- ✓ Know Ingress rules (host + path).
- ✓ Common tools: kubectl, Helm, Nginx, cloud providers.

DevOps Note

- ✓ Use Network Policies to control Pod-to-Pod traffic.
- ✓ Use CNI plugins (Calico, Cilium, Weave) for networking.
- ✓ Monitor services with Prometheus + Grafana.
- ✓ Check logs using kubectl logs / describe.
- ✓ Ingress + Cert-Manager for TLS.

Common KuweckI Commands

```
kubectl get pods -A          # all pods
kubectl get svc              # services
kubectl get ingress          # ingress
kubectl describe svc <name>  # details
kubectl logs <pod>           # pod logs
kubectl exec -it <pod> -- /bin/sh   # shell
kubectl port-forward <pod> 8080:80  # Forward
```

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Cloud Networking (AWS VPC)

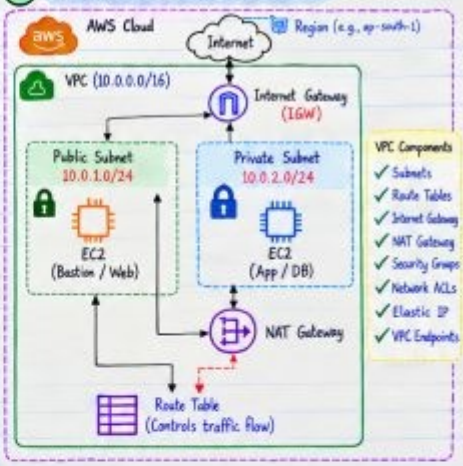
1 What is VPC ?

- VPC (Virtual Private Cloud) is a logically isolated network in AWS Cloud.
- We can define our own IP range (CIDR), subnets, route tables, gateways, security rules etc.
- It provides a secure and scalable network environment for our applications.

Key Point

✓ VPC is your own private network in AWS Cloud, isolated from other AWS networks.

2 VPC Architecture Overview



3 Public vs Private Subnets

Public Subnet

- Directly connected to Internet (via IGW)
- Used for resources like Web Server, Bastion Host, Load Balancer.
- Has route to 0.0.0.0/0 → Internet Gateway.

Private Subnet

- Not directly accessible from Internet.
- Used for App servers, Databases, internal services.
- Uses NAT Gateway for outbound Internet access (if required).

4 Internet Gateway (IGW)

- Connects VPC to the Internet.
- Allows two-way communication (for public subnets).
- VPC must be attached to the IGW.
- Used for resources like EC2 (public), ALB, etc.

5 NAT Gateway

- Allows instances in private subnets to access the Internet (outbound only).
- Provides Internet access without exposing instances to the Internet.
- Highly available within an AZ.
- Uses Elastic IP.
- Alternative: NAT Instance (manual setup).

6 Route Tables

- Define where network traffic should go.
- Each subnet is associated with a route table.
- Contains destination and target routes.
- Main components: Local route + custom routes.

Destination	Target	Purpose
10.0.0.0/16	local	VPC internal communication
0.0.0.0/0	igw-xxxx	Internet access (Public Subnet)
0.0.0.0/0	nat-xxxx	Outbound Internet (Private Subnet)

7 Security Groups vs NACLs

Feature	Security Group (SG)	Network ACL (NACL)
Type	Stateful (Allow only)	Stateless (Allow/Deny)
Level	Instance / ENI level	Subnet level
Rules	Only Allow rules	Allow and Deny rules
Traffic	Return traffic automatically allowed	Must allow both inbound and outbound
Order	No rule number (all evaluated)	Evaluated in order (lowest first)
Use Case	Control traffic to EC2, ALB, etc.	Additional subnet-level security layer

8 Best Practices

- ✓ Use separate public and private subnets.
- ✓ Use least privilege in Security Groups and NACLs.
- ✓ Use NAT Gateway for private subnet Internet access.
- ✓ Use VPC Endpoints for AWS services (S3, DynamoDB).
- ✓ Enable Flow Logs for monitoring and troubleshooting.
- ✓ Use multiple AZs for high availability.
- ✓ Keep CIDR planning in advance (avoid overlapping).
- ✓ Use tagging and proper naming.

9 Common Use Cases

- Host web applications (public subnet + ALB).
- Run application and database (private subnet).
- Connect on-premises to AWS using VPN / Direct Connect.
- Use VPC Endpoints for S3, DynamoDB (private access).
- Deploy EKS/ECS with public & private subnets.
- Build secure, isolated environments for Dev / Test / Prod.

10 AWS CLI Commands

```
aws ec2 create-vpc \
  --cidr-block 10.0.0.0/16                          # Create VPC

aws ec2 create-subnet \
  --vpc-id vpc-xxxxxxxxx \
  --cidr-block 10.0.1.0/24 \
  --availability-zone ap-south-1a                   # Create Subnet

aws ec2 create-internet-gateway                     # Create IGW

aws ec2 attach-internet-gateway \
  --vpc-id vpc-xxxxxxxxx \
  --internet-gateway-id igw-xxxxx                   # Attach IGW

aws ec2 create-nat-gateway \
  --subnet-id subnet-xxxxxxxxx \
  --allocation-id eipalloc-xxxxx                    # Create NAT GW

aws ec2 create-route-table \
  --vpc-id vpc-xxxxxxxxx                            # Create Route Table
```

11 Interview Tips

- Explain VPC, Subnet, IGW, NAT in simple words with diagram.
- Difference between Public and Private subnet.
- Difference between Security Group and NACL.
- When to use NAT Gateway vs NAT Instance?
- What is VPC Endpoint? Why use it?
- How route table works with examples?
- How to make an EC2 instance private?
- How to connect on-premises to AWS (VPN)?
- High availability design for VPC architecture.

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Networking Cheat Sheet

1 Key Comparisons

OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Function
7. Application	Application	User services and protocols
6. Presentation		
5. Session		
4. Transport	Transport	End-to-end communication
3. Network	Internet	Logical addressing and routing
2. Data Link	Network Access	Framing, MAC address, error detection
1. Physical		Transmits raw bits over medium

2 TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (ACK, retransmission)	Not reliable (no ACK)
Order	Maintains packet order	No order guarantee
Speed	Slower	Faster
Header Size	20 bytes	8 bytes
Use Case	Web, Email, File transfer, DB	Video/Voice Streaming, DNS, Gaming
Examples	HTTP, HTTPS, FTP, SSH	DNS, DHCP, VoIP, SNMP

3 HTTP vs HTTPS

Feature	HTTP	HTTPS
Full Form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Port	80	443
Security	Not encrypted	Encrypted (SSL/TLS)
Data	Plain text	Encrypted
Authentication	No	Yes (via SSL/TLS)
Use Case	Internal / Non-sensitive data	Secure / Sensitive data (banking, login, etc.)

Use HTTP for non-sensitive data and HTTPS for secure communication.

4 Common Network Commands

Command	Purpose	Example	Notes & Uses
ping	Test reachability and latency.	ping example.com (Windows: ping example.com)	Checks if host is reachable, shows time (latency) and packet loss.
traceroute	Trace path packets take to reach destination.	traceroute example.com (Windows: tracert example.com)	Shows all hops (routers) and latency at each hop.
nslookup	Query DNS to get domain information.	nslookup example.com	Get IP address, DNS server and other DNS records.
dig	Advanced DNS lookup tool (detailed info).	dig example.com (e.g. dig A example.com)	Detailed DNS records (A, MX, NS, etc.).
curl	Test web calls (HTTP/HTTPS), view response headers.	curl -I https://example.com curl -X GET https://api.example.com	Test API, headers, response code and response data.
ip	Show / configure network interfaces and IP addresses.	ip addr ip route	View IP, interface, routes. (Linux)
ss	Show socket statistics and listening ports.	ss -tuln ss -tnp	List open ports, active connections, process info.

5 Common Ports (Most Used)

Port	Service	Protocol	Use Case
20, 21	FTP	TCP	File transfer
22	SSH	TCP	Remote login
23	Telnet	TCP	Remote login (insecure)
25	SMTP	TCP	Email sending
53	DNS	UDP/TCP	Domain name resolution
80	HTTP	TCP	Web traffic (non-secure)
443	HTTPS	TCP	Secure web traffic
3306	MySQL	TCP	MySQL database
5432	PostgreSQL	TCP	PostgreSQL database
6379	Redis	TCP	Redis cache
27017	MongoDB	TCP	MongoDB database

6 Key Reminders

- DNS translates domain names to IP addresses.
- IP is used to identify a device.
- TCP is reliable, UDP is faster.
- Use HTTPS for secure data.
- Always use NAT for private subnets.
- Firewalls and security groups control traffic.
- Monitor network traffic and logs.
- Use proper ports and protocols.
- Keep network design simple and scalable.

7 Best Practices

- Use least privilege access.
- Keep services up to date.
- Use encrypted protocols (HTTPS, SSH).
- Use firewalls / security groups.
- Monitor network traffic and logs.
- Implement network segmentation (VPC, subnets).
- Use private subnets for internal services.
- Enable MFA and strong passwords.
- Regularly review and remove unused open ports.
- Follow security best practices (AWS, on-prem, cloud).

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